

Paper Summary:

RenderDiffusion: Image Diffusion for 3D Reconstruction, Inpainting and Generation

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5 MAY, 2025

Citation

Titas Anciukevičius, Zexiang Xu, Matthew Fisher, Paul Henderson, Hakan Bilen, Niloy J. Mitra, Paul Guerrero. *RenderDiffusion: Image Diffusion for 3D Reconstruction, Inpainting and Generation*. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, pp. 12608–12618. https://openaccess.thecvf.com/content/CVPR2023/html/Anciukevicius_RenderDiffusion_Image_Diffusion_for_3D_Reconstruction_Inpainting_and_Generation_CVPR_2023_paper.html

1 Problem and Motivation

Image diffusion models were, by 2023, state of the art for 2D image generation, but had not been successfully extended to 3D tasks such as view-consistent 3D generation or single-view 3D reconstruction. Prior attempts to build 3D diffusion models either (1) operated directly on explicit 3D data such as point clouds, voxels, or SDFs, which requires costly 3D-supervised training data (3D model repositories are orders of magnitude smaller than image datasets) and incurs heavy memory cost, or (2) optimized a NeRF using a pretrained 2D diffusion model as a prior (e.g., DreamFusion), which is slow (per-scene optimization) and does not yield a true generative model that can be sampled directly.

RenderDiffusion asks whether a diffusion model can learn 3D structure while being trained *only* on 2D images (with known camera poses) and no explicit 3D supervision.

2 Method

2.1 Core Idea

Standard 2D diffusion models train a UNet denoiser g_θ to iteratively remove noise from an image. RenderDiffusion keeps this same training/sampling framework but changes the *architecture* of the denoiser itself: instead of directly predicting a denoised 2D image, the denoiser first reconstructs an explicit 3D scene representation from the noisy input image, then volumetrically renders that 3D representation back into a 2D image. Because the network is forced to pass through a 3D representation at every denoising step, 3D structure emerges purely from 2D supervision.

2.2 Architecture

The denoiser is factored as

$$g_{\theta}(\mathbf{x}_t, t, \mathbf{v}) := r_{\psi}(e_{\phi}(\mathbf{x}_t, t), \mathbf{v}),$$

consisting of:

- **Triplane encoder** e_{ϕ} . A UNet-based encoder maps the noisy input image $\mathbf{x}_t$ (with known camera view $\mathbf{v}$) to a *triplane* representation P : three orthogonal 2D feature maps that together approximate a 3D feature volume compactly (avoiding the cubic memory cost of a full voxel grid).
- **Triplane renderer** r_{ψ} . A small MLP s_{ψ} converts the triplane feature at any queried 3D point p (obtained via bilinear interpolation and summation across the three planes, denoted $P[p]$) into a density γ and color c . These are integrated along camera rays using standard volumetric rendering (MipNeRF-style, two-pass importance sampling as in NeRF) to produce the output image $\tilde{\mathbf{x}}_0$.

Unlike diffusion models trained on a pretrained latent space (e.g., Latent Diffusion Models), this model works directly in 2D pixel space, which the authors argue yields sharper results. Following standard DDPM training, but adapted for the 3D reconstruction sub-task, the network is trained to predict the clean image $\mathbf{x}_0$ (rather than the noise ϵ) directly:

$$\mathcal{L} := \|g_{\theta}(\mathbf{x}_t, t) - \mathbf{x}_0\|_1.$$

This means the triplane produced at each denoising step is a genuine (if partial/noisy) estimate of the underlying 3D scene, which becomes fully resolved 3D content by the final step and can be rendered from novel viewpoints.

2.3 Score-Distillation Regularization

On face datasets (FFHQ/AFHQ), naively training this way can collapse to “trivial” geometry (e.g., a flat billboard) since only the input viewpoint is directly supervised. To prevent this, the authors add a score-distillation loss (inspired by DreamFusion): at each training step, the reconstructed 3D scene is additionally rendered from a random viewpoint, and a diffusion loss is applied there too, encouraging plausibility from all angles, not just the observed view.

2.4 Downstream Capabilities from One Trained Model

- **Monocular 3D reconstruction.** An input image is noised for t_r steps and denoised back; the intermediate triplane is the reconstructed 3D scene, renderable from novel views. $t_r=0$ gives high-fidelity in-distribution reconstruction; larger t_r trades fidelity for robustness to out-of-distribution inputs, since the denoiser can “correct” atypical inputs toward the learned data distribution.
- **Unconditional 3D generation.** Standard ancestral sampling from noise produces a novel 3D scene (via its triplane), which can then be rendered from arbitrary views.
- **3D-aware inpainting.** Following the RePaint approach, known pixel regions are re-noised and clamped at each reverse step while unknown/masked regions are sampled normally, yielding a 3D scene whose masked region is filled in a way consistent with the visible pixels and plausible in 3D.

3 Experimental Setup

Evaluated on four datasets: FFHQ (real faces) and AFHQv2 (cat faces), both using camera poses estimated as in EG3D; and two synthetic datasets, CLEVR1 (single simple object per scene, 900 scenes) and three ShapeNet categories (car, plane, chair; 3200 objects each), rendered from 100 viewpoints per object using Blender. No explicit 3D supervision is used anywhere – only posed 2D renderings.

4 Results

4.1 Monocular 3D Reconstruction (Table 1)

Method	car		plane		chair		average		CL
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
PixelNeRF (multi-view sup.)	27.3	.838	29.4	.887	30.2	.884	28.9	.870	43.4
EG3D (single-view sup.)	21.8	.714	25.0	.803	25.5	.803	24.1	.773	33.2
RenderDiffusion (single-view sup.)	25.4	.805	26.3	.834	26.6	.830	26.1	.823	39.8

Table 1: RenderDiffusion outperforms EG3D (the only other method trained with the same single-view-only supervision) on every dataset. PixelNeRF, trained with multi-view supervision (an unfair advantage), is shown as an upper-reference and is not directly comparable.

Qualitatively, EG3D (via GAN inversion) often produces shapes/colors that visibly deviate from the input image; PixelNeRF reconstructs the input view faithfully but becomes blurry from novel viewpoints; RenderDiffusion strikes a balance, preserving input identity while remaining sharp from novel views. It is also far faster at inference: reconstruction takes under 0.03 s/scene versus roughly 3 minutes for EG3D’s optimization-based inversion.

4.2 Unconditional Generation (Table 2)

Measured by dataset coverage (higher is better): Both methods produce realistic, 3D-consistent

Method	car	plane	chair	FFHQ	AFHQ
GIRAFFE	0.11	0.45	0.39	0.66	0.07
EG3D	0.70	0.61	0.71	0.68	0.37
RenderDiffusion	0.46	0.84	0.85	0.31	0.26

Table 2: RenderDiffusion beats EG3D/GIRAFFE on two of three ShapeNet categories despite being trained purely by denoising, with somewhat lower coverage on FFHQ/AFHQ.

scenes with plausible shadows, but RenderDiffusion shows somewhat higher sample diversity in color and shape.

4.3 3D-Aware Inpainting

Qualitative results show the model can fill masked 2D regions with diverse, plausible completions that remain consistent in 3D across novel viewpoints – a task the model was never explicitly trained for.

5 Contributions

1. First diffusion model for 3D generation and inference trained purely from monocular (single-view) 2D images with no explicit 3D supervision.
2. A novel denoiser architecture that embeds an explicit 3D representation (triplane + volumetric renderer) directly inside the 2D diffusion denoising step, creating a strong inductive bias toward 3D-consistent outputs.
3. A single trained model supports three tasks without task-specific retraining: monocular 3D reconstruction, unconditional 3D generation, and 3D-aware inpainting/editing.
4. Competitive or superior performance versus prior single-view-supervised methods (EG3D), approaching methods trained with the unfair advantage of multi-view supervision (PixelNeRF), while being orders of magnitude faster than optimization-based reconstruction (GAN inversion).

6 Limitations

- Generated images are somewhat blurrier than GAN-based methods in some cases.
- The score-distillation regularization needed to avoid trivial/flat geometry on face datasets reduces fidelity of generated 3D shapes.
- Requires known camera extrinsics at training time, and triplanes are anchored to a global coordinate frame, limiting generalization across object placement.
- Does not yet support fine-grained object/material editing.

7 Take-away

RenderDiffusion’s core contribution is architectural: rather than diffusing directly in 3D (which needs scarce 3D data) or using a 2D diffusion prior to optimize a NeRF per-scene (slow, not a true generative model), it builds 3D structure *into* the denoiser of a standard 2D image diffusion model via a triplane representation and differentiable volume rendering, trained end to end with only posed 2D images and an L_1 image reconstruction loss. This inductive bias is enough to obtain a genuine 3D generative/inference model – usable for reconstruction, generation, and inpainting – without ever supervising with 3D geometry directly.